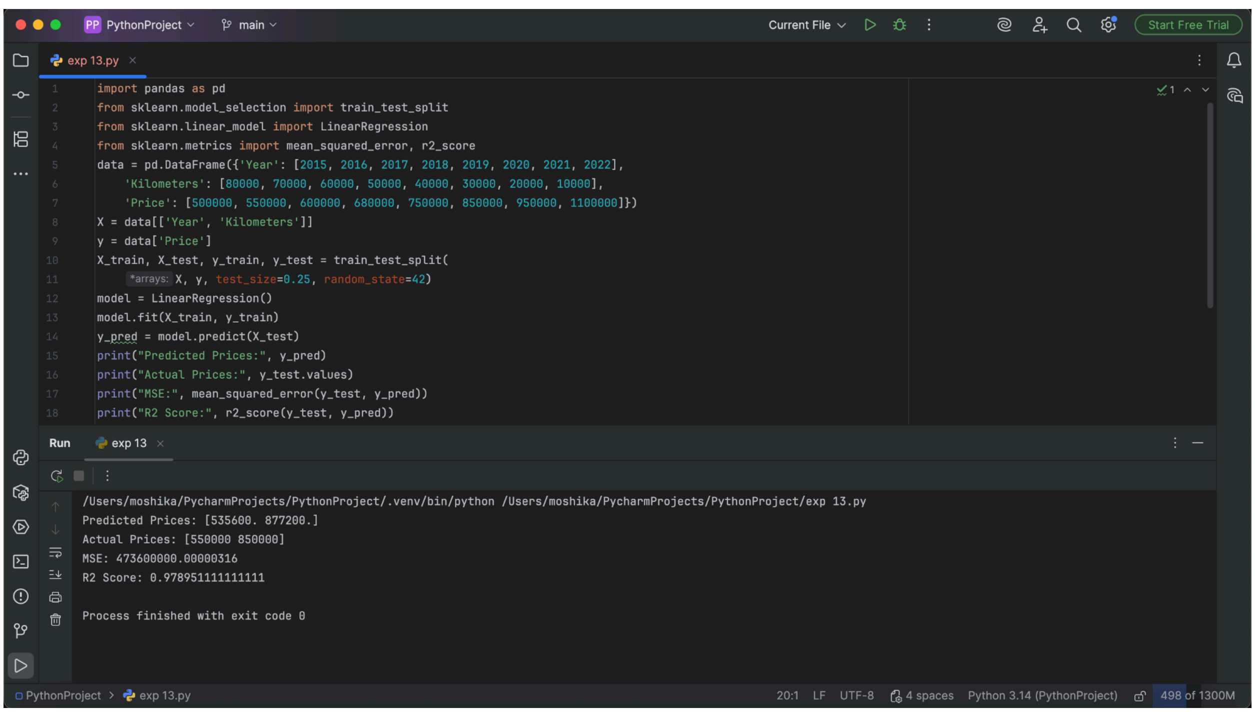
EXPERIMENT - 13

Aim: To predict the price of a car using a suitable machine learning algorithm.

Procedure:

1. Load the car price dataset.
2. Clean and preprocess the data.
3. Select relevant features such as year, mileage, and engine capacity.
4. Separate input and target variables.
5. Split the dataset into training and testing sets.
6. Train a regression model.
7. Predict car prices for the test data.
8. Evaluate the model using an appropriate error metric.
9. Display the predicted prices.

Output:



```python
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
data = pd.DataFrame({'Year': [2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022],
    'Kilometers': [80000, 70000, 60000, 50000, 40000, 30000, 20000, 10000],
    'Price': [500000, 550000, 600000, 680000, 750000, 850000, 950000, 1100000]})
X = data[['Year', 'Kilometers']]
y = data['Price']
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, random_state=42)
model = LinearRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
print("Predicted Prices:", y_pred)
print("Actual Prices:", y_test.values)
print("MSE:", mean_squared_error(y_test, y_pred))
print("R2 Score:", r2_score(y_test, y_pred))
```

```
/Users/moshika/PycharmProjects/PythonProject/.venv/bin/python /Users/moshika/PycharmProjects/PythonProject/exp 13.py
Predicted Prices: [535600. 877200.]
Actual Prices: [550000 850000]
MSE: 473600000.00000316
R2 Score: 0.978951111111111

Process finished with exit code 0
```

Result: The model successfully predicted car prices based on the given car features.